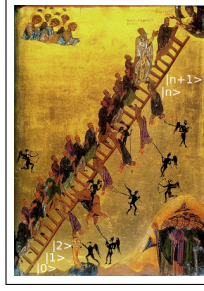


PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026



Harmonic Oscillator [Wikimedia]

EXERCISE 4.1: CONVOLUTION PRODUCT (6P)

Let us consider the probability density of the gaps between Poisson-distributed events (the length x of silence between the clicks of a Geiger counter), which is defined by

$$p(x) = we^{-wx} \quad (x \geq 0, w > 0).$$

- (a) Compute the corresponding characteristic function (see lecture notes). (1P)
- (b) Compute the two-fold convolution product $(p * p)_{(x)}$ and the three-fold convolution product $(p * p * p)_{(x)}$. (2P)
- (c) Guess a general formula for the N -fold convolution product $p_{(x)}^{*N}$ and prove it by recursion. (2P)
- (d) Fit a normal distribution with mean μ and width σ to the probability density computed in (c) for large N . *Hint: Recall the behavior of cumulants* (1P)

EXERCISE 4.2: HARMONIC OSCILLATOR (6P)

Consider a harmonic oscillator with eigenstates $|n\rangle = |0\rangle, |1\rangle, |2\rangle, \dots$ and eigenenergies $E_n = \hbar\omega(n + \frac{1}{2})$. Let us assume that the eigenstates correspond to classical configurations and that the system evolves stochastically according to a continuous-time Markov process as follows:

$$\begin{aligned} n &\rightarrow n + 1 && \text{with rate } w_{\uparrow} \\ n &\rightarrow n - 1 && \text{with rate } w_{\downarrow} \end{aligned}$$

- (a) Calculate the Liouville operator $\mathcal{L}$. (1P)
- (b) Determine the left and the right eigenvector to the eigenvalue zero. (2P)
- (c) Normalize the right eigenvector obtained in (b) so that it can be interpreted as a probability distribution. Under which condition is the normalization possible? (2P)
- (d) In a canonical ensemble the probability of finding the system at energy E_n is proportional to $\exp(-\frac{E_n}{k_B T})$, where T is the temperature and k_B is the Boltzmann constant. Express the temperature T in terms of the rates $w_{\uparrow}$ and $w_{\downarrow}$. (1P)

($\Sigma = 12P$)

Please submit your solution electronically as a single PDF file until Wednesday, May 13, 2026, via Wue-Campus.